Notes on: Dynamical Low Rank approximation of PDEs with random parameters

Stefan Ehin

June 22, 2026

Problem

1. Regular PDE problem statements do not account for the potential error in many parameters: coefficients of the operator, forcing/source terms, initial conditions and boundary conditions.
2. (a) Monte Carlo method is simple but has slow convergence in sample size.
   (b) Spectral methods are considerably more compute-efficient (assuming the underlying solution map is smooth) but it become compute-heavy when the stochastic space is high-dimensinal.
   (c) In case the solution manifold is approximately linear the curse of dimensionality is broken but it fails for manifolds that strongly dimiate from being linear.

Key idea

1. Model all the error-prone variables as probability distributions.
2. Use a dynamical low-rank basis i.e. reduced basis like in 2. (C) but evolving over time.

What I'd prove/extend